Project Milestone 4

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Load packages and datasets

```
library(tidyverse)

d3_instLevel = read_csv("../clean_data/d3_instLevel_ds.csv")
d3_sports = read_csv("../clean_data/d3_sports_ds.csv")
d3_coach = read_csv("../clean_data/d3_coach_ds.csv")


nwc = c( # NWC teams with the exception of Willamette
  "Linfield University", "George Fox University", "Whitworth University",
  "Whitman College", "Pacific University", "University of Puget Sound",
  "Pacific Lutheran University", "Lewis & Clark College"
)
```

EDA plots

Plot 1: Operating Expenses vs Number of Athletes

- Geometries: Points, ABline, text
- Aesthetics: x = number of athletes, y = operating expenses, color = type of school

```
selected_sport = "Men's Football" #User chooses
selected_year = 2025 #User chooses

title = paste0("Willamette ", #clean title that updates with selections
               selected_sport,
               " Spends Less In Comparison to Other NWC Schools in ",
               as.numeric(selected_year))

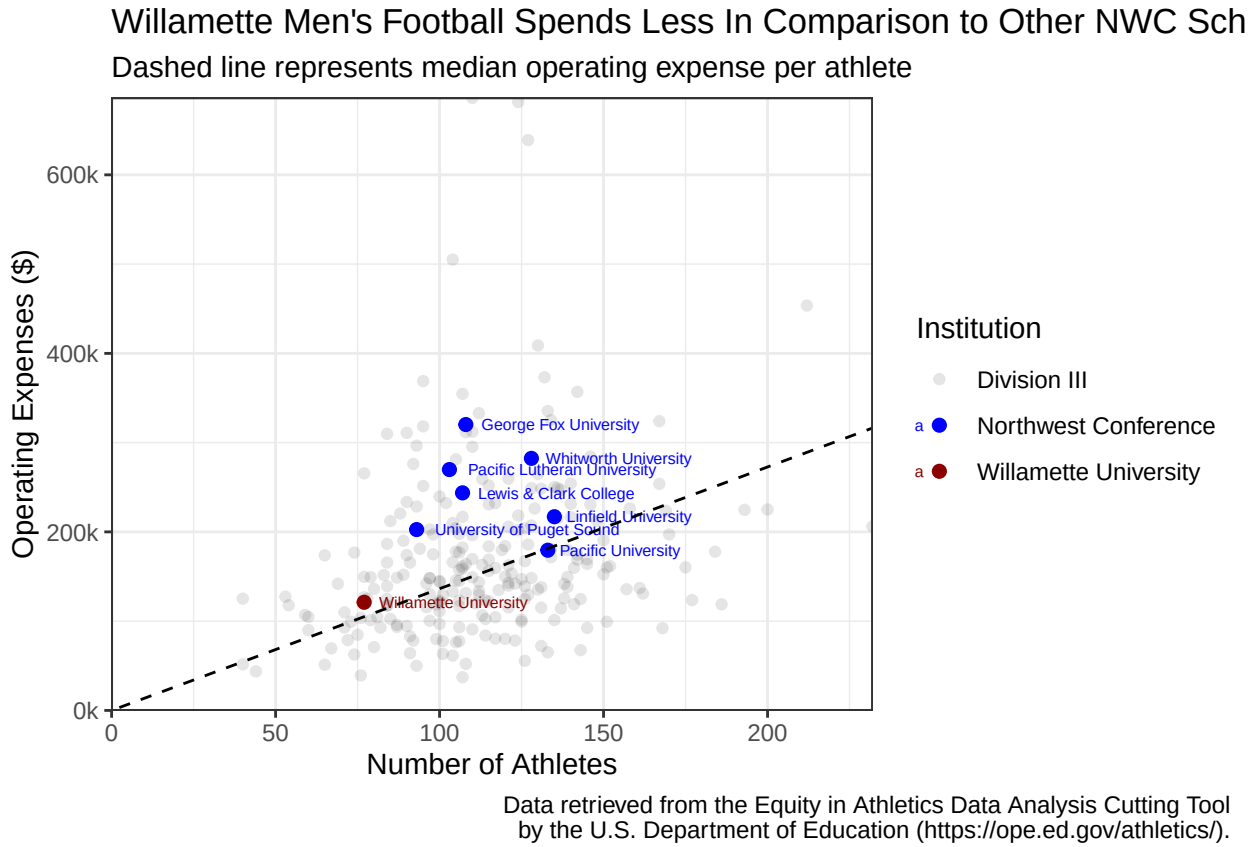
ds = d3_sports %>%
  filter(sport == selected_sport, year == selected_year) %>%
  mutate(
    color = case_when(
      institution_name %in% nwc ~ "Northwest Conference",
      institution_name == "Willamette University" ~ "Willamette University",
      .default = "Division III"
    ) #This is for later grouping
```

```
  )

colorpal = c( # color palette to be used in graph
  "Willamette University" = "darkred",
  "Northwest Conference" = "blue",
  "Division III" = "black"
)


solid = ds %>% # all labeled points on the plot
  filter(institution_name %in% nwc | institution_name == "Willamette University")


ds %>%
  filter(!(institution_name %in% nwc), # all other d3 points on the plot
         !(institution_name == "Willamette University")) %>%
  ggplot(aes(x = n_athletes, y = op_exp, color = color)) +
  geom_point(alpha = .1) + # d3 points
  geom_point(data = solid, # nwc points
             aes(x = n_athletes, y = op_exp, color = color),
             size = 2) +
  geom_text(data = solid, # labels for nwc points
            aes(label = institution_name),
            hjust = -.1,
            size = 2) +
  geom_abline(intercept = 0, # black dashed median line
              slope = median(ds$op_exp, na.rm = TRUE)/median(ds$n_athletes),
              color = "black",
              linetype = "dashed") +
  scale_color_manual(values = colorpal) + # add color palette
  scale_x_continuous(limits = c(0, max(ds$n_athletes)), expand = c(0, 0)) +
  scale_y_continuous(labels = function (x) paste0(x/1000, "k"),
                     limits = c(0, max(ds$op_exp)),
                     expand = c(0, 0)) +
  theme_bw() +
  theme(
    plot.caption.position = "plot" #Moving caption to the right side
  ) +
  labs(
    title = title,
    subtitle = "Dashed line represents median operating expense per athlete",
    caption = "Data retrieved from the Equity in Athletics Data Analysis Cutting Tool\nby the U.S. Depa
    x = "Number of Athletes",
    y = "Operating Expenses ($)",
    color = "Institution"
  )
```



Analysis: This is our main point, that Willamette spends less than other schools, which leads us with less opportunities for growth. By presenting this graphic, which can show multiple different sports and years, we can show that this finding is consistent and change needs to be made to increase the competitiveness and success of our sports teams.

Plot 2: Operating Expenses per Athlete by Year

- Geometries: Points, smooth line
- Aesthetics: x = year, y = operating expenses per athlete, color = type of school

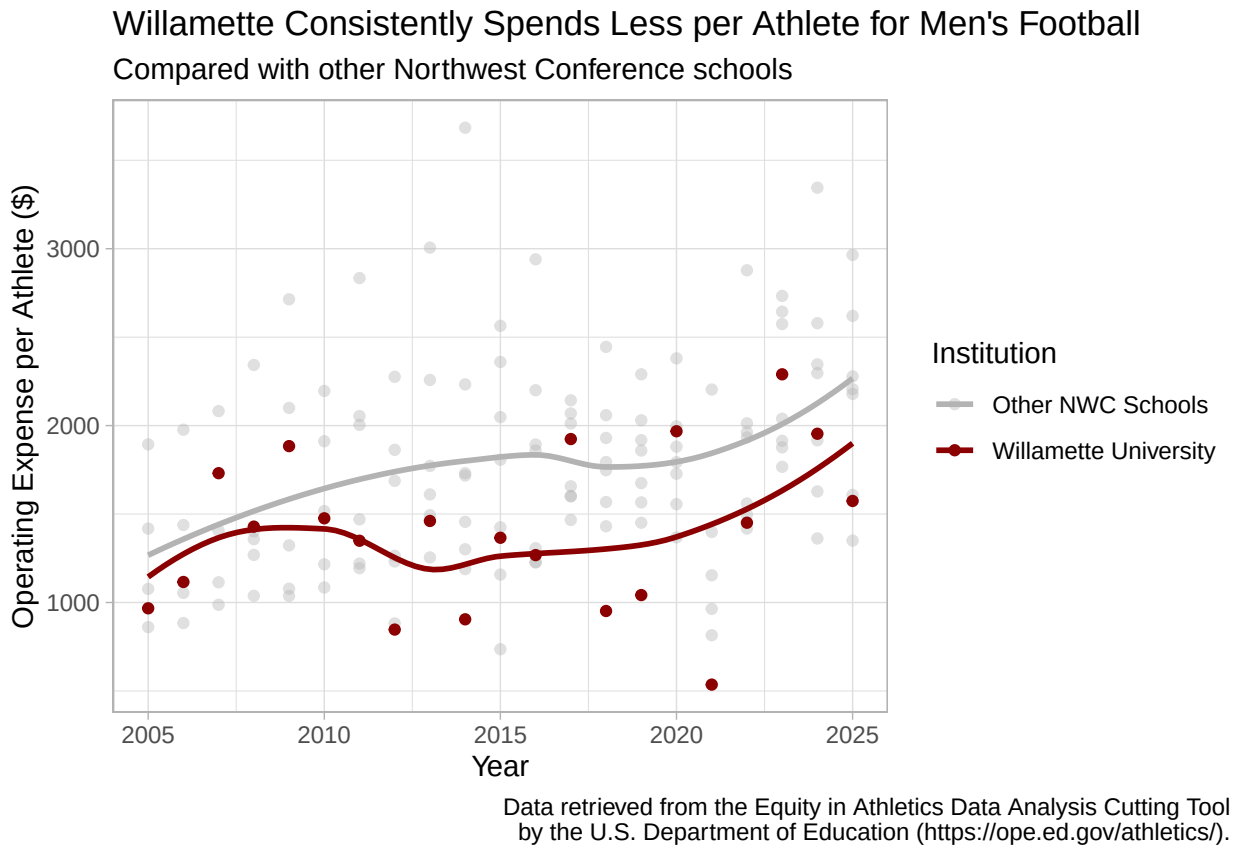
```
selected_sport = "Men's Football" #User chooses

# clean title for plot that updates with user selections
title = paste0("Willamette Consistently Spends Less per Athlete for ", selected_sport)

# other nwc points
points = d3_sports %>%
  filter(sport == selected_sport, institution_name %in% nwc)

# willamette points
willamette = d3_sports %>%
  filter(sport == selected_sport, institution_name == "Willamette University")
```

```r
points %>%
  ggplot(aes(x = year, y = op_exp_per_athlete)) +
  geom_point(alpha = .4, aes(color = "Other NWC Schools")) + # nwc points
  geom_smooth(se = F, alpha = .5, aes(color = "Other NWC Schools")) + # nwc line
  geom_point(data = willamette, # willamette points
             aes(x = year, y = op_exp_per_athlete, color = "Willamette University")) +
  geom_smooth(data = willamette, # willamette line
              se = F,
              aes(color = "Willamette University")) +
  theme_light() +
  theme(
    plot.caption.position = "plot" #Moving caption to the right side
  ) +
  scale_color_manual( # adds different colors for nwc vs willamette geoms
    values = c(
      "Willamette University" = "darkred",
      "Other NWC Schools" = "grey70"
    )
  ) +
  labs(
    title = title,
    subtitle = "Compared with other Northwest Conference schools",
    x = "Year",
    y = "Operating Expense per Athlete ($)",
    color = "Institution",
    caption = "Data retrieved from the Equity in Athletics Data Analysis Cutting Tool\nby the U.S. Depa
  )
```



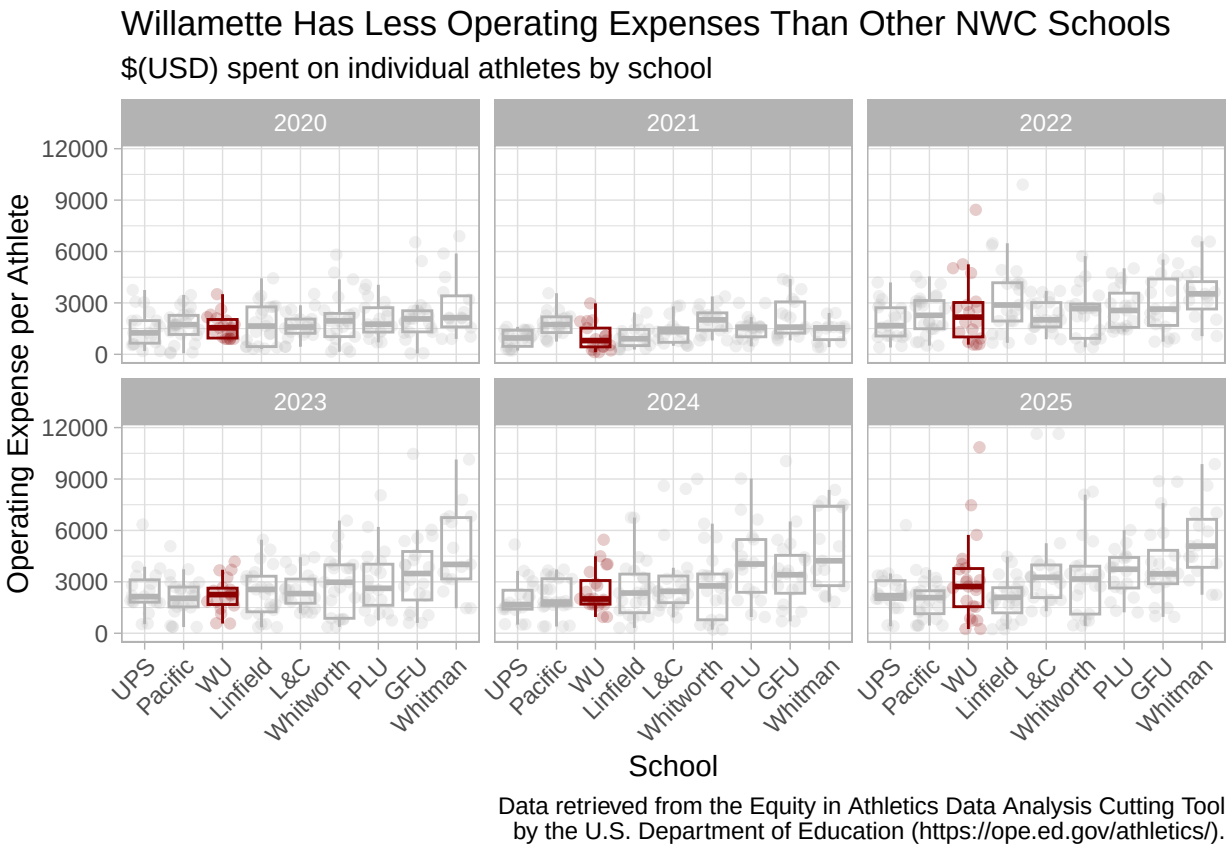
Analysis: This also demonstrates our main point, that Willamette spends less than other schools, which leads us with less opportunities for growth. By presenting this graphic which can show multiple different sports over time, we can show that this finding is consistent and change needs to be made, allowing our sports to be more competitive and successful.

Plot 3: Operating Expenses per Athlete by School

- Geometries: Points, boxplot
- Aesthetics: x = school, color = type of school

```
d3_sports %>%
  filter(institution_name %in% nwc | institution_name == "Willamette University",
         year >= 2020) %>% # only include nwc and 2020 and later rows
  ggplot(aes(x = fct_reorder(institution_name, op_exp_per_athlete, .fun = median),
             y = op_exp_per_athlete)) +
  geom_point(aes(color = institution_name == "Willamette University",
                 group = institution_name),
             alpha = .2, position = "jitter") + # scatterplot under boxplots
  geom_boxplot(aes(color = institution_name == "Willamette University",
                   group = institution_name), alpha = 0) + # overlaid boxplots
  facet_wrap(~year) + #Comparing between years
  theme_light() +
  theme(
    plot.caption.position = "plot", #Moving caption to the right side
    axis.text.x = element_text(angle = 45, hjust = 1) # angle x labels
```

```
)  +
scale_x_discrete(
  labels = c( # shorten names to make labels more readable
    "Willamette University" = "WU", "University of Puget Sound" = "UPS",
    "Lewis & Clark College" = "L&C", "Pacific Lutheran University" = "PLU",
    "Pacific University" = "Pacific", "Whitworth University" = "Whitworth",
    "George Fox University" = "GFU", "Linfield University" = "Linfield",
    "Whitman College" = "Whitman"
  )
) +
scale_color_manual( # add coloring for nwc vs willamette
  values = c("TRUE" = "darkred", "FALSE" = "gray70"),
  guide = "none" # no legend
) +
labs(
  x = "School",
  y = "Operating Expense per Athlete",
  title = "Willamette Has Less Operating Expenses Than Other NWC Schools",
  subtitle = "$(USD) spent on individual athletes by school",
  caption = "Data retrieved from the Equity in Athletics Data Analysis Cutting Tool\nby the U.S. Depa
)
```



Analysis: This also demonstrates our main point, that Willamette spends less than other schools, which leads us with less opportunities for growth. Even across multiple years, Willamette ranks near the bottom. By presenting this graphic which shows data for all sports at once across multiple different years, we can show that this finding is consistent and change needs to be made, allowing our sports teams to be more

competitive and successful.

```
library(tidytext)


school_labels = c(
  "Willamette University" = "WU", "University of Puget Sound" = "UPS",
  "Lewis & Clark College" = "L&C", "Pacific Lutheran University" = "PLU",
  "Pacific University" = "Pacific", "Whitworth University" = "Whitworth",
  "George Fox University" = "GFU", "Linfield University" = "Linfield",
  "Whitman College" = "Whitman"
)

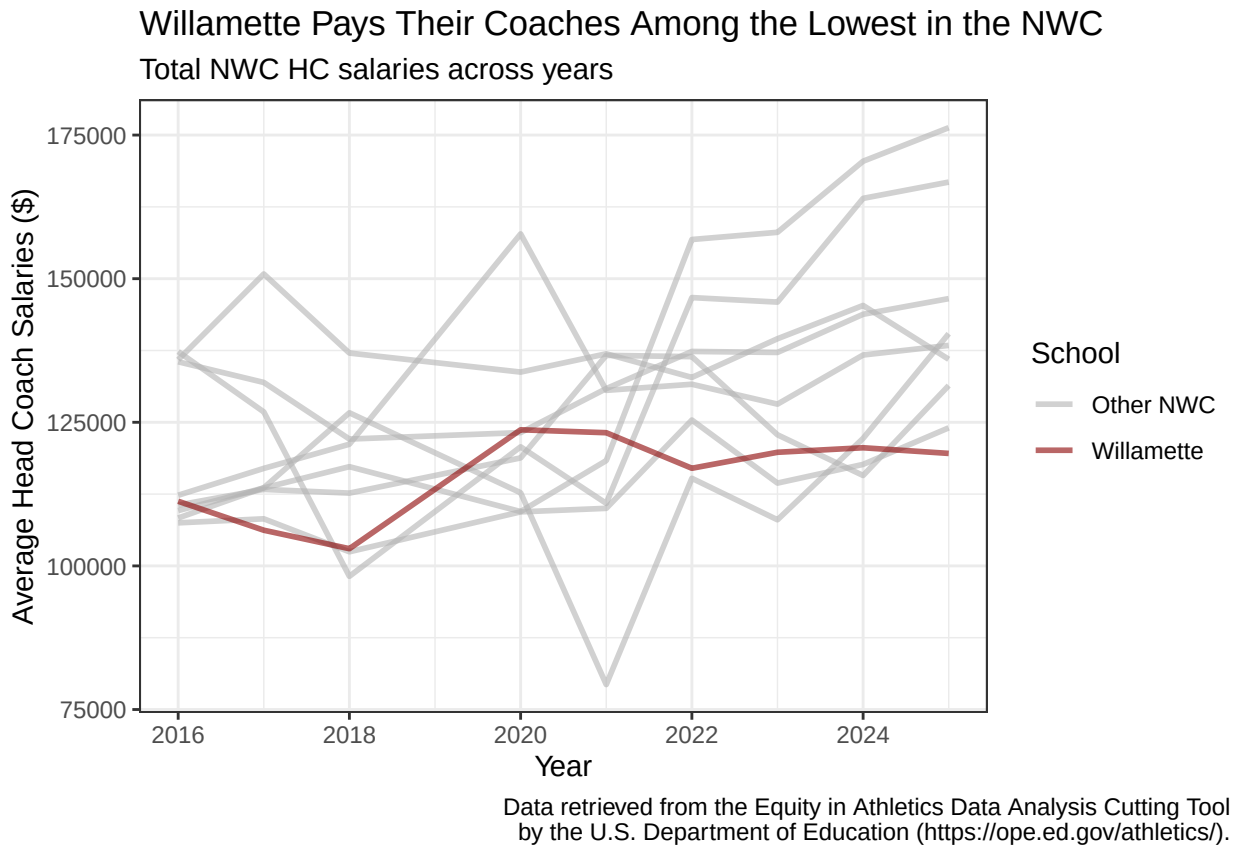

d3_sports %>%
  filter(institution_name %in% nwc | institution_name == "Willamette University", year == 2025, sport %
  ggplot(aes(x = reorder_within(institution_name, total_hc + total_ac, sport), y = total_hc + total_ac)
  geom_col(aes(fill = institution_name == "Willamette University",
        group = institution_name)) +
  theme_light() +
  facet_wrap(
    ~ sport,
    scales = "free_x",
    labeller = as_labeller(c(
      "Women's Basketball" = "WBB",
      "Men's Basketball" = "MBB",
      "Women's Soccer" = "WSOC",
      "Men's Soccer" = "MSOC",
      "Women's Softball" = "Softball",
      "Men's Baseball" = "Baseball",
      "Men's Football" = "Football",
      "Women's Volleyball" = "Volleyball",
      "Men's Track and Field (Indoor)" = "M T&F (I)",
      "Men's Track and Field (Outdoor)" = "M T&F (O)",
      "Women's Track and Field (Indoor)" = "W T&F (I)",
      "Women's Track and Field (Outdoor)" = "W T&F (O)",
      "Women's Lacrosse" = "Lacrosse",
      "Men's Cross Country" = "M XC",
      "Women's Cross Country" = "W XC",
      "Men's Golf" = "M Golf",
      "Women's Golf" = "W Golf",
      "Men's Swimming" = "M Swim",
      "Women's Swimming" = "W Swim",
      "Women's Tennis" = "W Tennis",
      "Men's Tennis" = "M Tennis",
      "Women's Other Sports" = "W Other Sports"
    ))
  ) +
  scale_x_reordered(
    breaks = c()
  ) +
  scale_fill_manual(
    values = c("TRUE" = "darkred", "FALSE" = "gray70"),
    labels = c("TRUE" = "Willamette", "FALSE" = "Other NWC")
```

```
  ) +
  labs(
    x = "",
    y = "Total Number of Coaches",
    fill = "School",
    title = "Is Willamette Undercoached?",
    subtitle = "Total # of coaches by sport in the NWC"
  )
```

Plot 4: Headcoach Salary vs Year

- Geometries: Lines
- Aesthetics: x = year, y = head coach salaries, color = type of school

```
d3_instLevel %>%
  mutate(
    HC_sal = HDCOACH_SAL_FTE_MEN+HDCOACH_SAL_FTE_WOMN
  ) %>% #Adding averages of full time HC sal
  filter(institution_name %in% nwc| institution_name == "Willamette University")%>%
  # only include nwc teams
  ggplot(aes(x = year, y = HC_sal)) +
  geom_line(linewidth = 1,
            alpha = .6,
            aes(color = institution_name == "Willamette University",
                group = institution_name)) +
  scale_x_continuous(breaks = c(2016,2018,2020,2022,2024))+
  scale_color_manual( # add coloring for nwc vs willamette
    values = c("TRUE" = "darkred", "FALSE" = "gray70"),
    labels = c("TRUE" = "Willamette", "FALSE" = "Other NWC")
  )+
  theme_bw() +
  theme(
    plot.caption.position = "plot" #Moving caption to the right side
  ) +
  labs(
    title ="Willamette Pays Their Coaches Among the Lowest in the NWC",
    subtitle = "Total NWC HC salaries across years",
    y = "Average Head Coach Salaries ($)",
    x = "Year",
    color = "School",
    caption = "Data retrieved from the Equity in Athletics Data Analysis Cutting Tool\nby the U.S. Depa
  )
```



Analysis: Willamette, on average, per individual coach, ranks the lowest in 2025 in the Northwest Conference. This is significant because Willamette sports are continuing to generally struggle. Inability to pay top candidates hurts the ability of our teams to be the most competitive they can, because development of our athletes is limited. Investing higher amounts in coaches is needed to observe change instead of seeing a continuous slump.

```
d3_instLevel %>%
  mutate(HC_sal = HDCOACH_SAL_FTE_MEN+HDCOACH_SAL_FTE_WOMN) %>%
  filter(institution_name %in% nwc| institution_name == "Willamette University")%>%
  ggplot(aes(x = year, y = HC_sal, color = institution_name)) +
  geom_line(linewidth = 1, alpha = .6) +
  scale_x_continuous(breaks = c(2016,2018,2020,2022,2024))+
  # scale_color_manual(
  #   values = c("TRUE" = "darkred", "FALSE" = "gray70"),
  #   guide = "none"
  # )+
  theme_bw()


nwc_med = d3_instLevel %>%
  filter(institution_name %in% nwc) %>%
  summarize(
    .by = year,
    med_opexp_athlete = median(IL_TOTAL_OPEXP_INCLCOED/(UNDUP_CT_PARTIC_MEN + UNDUP_CT_PARTIC_WOMEN)),
    med_exp_athlete = median(GRND_TOTAL_EXPENSE/(UNDUP_CT_PARTIC_MEN + UNDUP_CT_PARTIC_WOMEN))
  )
```

```
wu_med = d3_instLevel %>%
  filter(institution_name == "Willamette University") %>%
  summarize(
    .by = year,
    med_opexp_athlete = median(IL_TOTAL_OPEXP_INCLCOED/(UNDUP_CT_PARTIC_MEN + UNDUP_CT_PARTIC_WOMEN)),
    med_exp_athlete = median(GRND_TOTAL_EXPENSE/(UNDUP_CT_PARTIC_MEN + UNDUP_CT_PARTIC_WOMEN))
    )


nwc_med %>%
  ggplot(aes(x = year)) +

  # Other NWC schools: operating expenses (dashed)
  geom_point(aes(y = med_opexp_athlete, color = "Other NWC Schools")) +
  geom_line(aes(y = med_opexp_athlete, color = "Other NWC Schools", linetype = "Operating Expenses")) +

  # Other NWC schools: total expenses (solid)
  geom_point(aes(y = med_exp_athlete, color = "Other NWC Schools")) +
  geom_line(aes(y = med_exp_athlete, color = "Other NWC Schools", linetype = "Total Expenses")
  ) +

  # Willamette: operating expenses (dashed)
  geom_point(data = wu_med, aes(y = med_opexp_athlete, color = "Willamette University")) +
  geom_line(data = wu_med, aes(y = med_opexp_athlete, color = "Willamette University", linetype = "Opera

  # Willamette: total expenses (solid)
  geom_point(data = wu_med, aes(y = med_exp_athlete, color = "Willamette University")) +
  geom_line(data = wu_med, aes(y = med_exp_athlete, color = "Willamette University", linetype = "Total

  scale_color_manual(
    name = "Institution",
    values = c(
      "Other NWC Schools" = "grey60",
      "Willamette University" = "darkred"
    )
  ) +
  scale_linetype_manual(
    name = "Expense Type",
    values = c(
      "Operating Expenses" = "dashed",
      "Total Expenses" = "solid"
    )
  ) +
  scale_x_continuous(
    breaks = seq(2000, 2025, by = 1)
  ) +
  theme_light() +
  labs(
    title = "Less of Willamette's Athletic Expenses Go Directly to Athletes",
    subtitle = "Compared with other NWC schools",
    x = "Year",
    y = "Expenses per Athlete ($)"
  )
```

```
nwc_ = d3_instLevel %>%
  filter(institution_name %in% nwc)

wu = d3_instLevel %>%
  filter(institution_name == "Willamette University")

nwc_ %>%
  bind_rows(wu) %>%
  ggplot(aes(x = year, y = RECRUITEXP_TOTAL/(UNDUP_CT_PARTIC_MEN + UNDUP_CT_PARTIC_WOMEN))) +
  geom_point(aes(color = institution_name == "Willamette University")) +
  geom_smooth(data = wu, se = F, color = "darkred") +
  geom_smooth(data = nwc_, se = F, color = "grey70") +
  theme_light() +
  scale_color_manual(
    values = c("TRUE" = "darkred", "FALSE" = "gray70"),
    labels = c("TRUE" = "Willamette", "FALSE" = "Other NWC")
  ) +
  scale_x_continuous(
    breaks = seq(2000, 2025, by = 1)
  ) +
  labs(
    title = "How Much is Spent on Recruiting Over Time",
    x = "Year",
    y = "Amount Spent on Recruiting per Athlete ($)",
    color = "Institution"
  )
```

Plot 5: Percent Difference in Expense Categories from NWC Median

- Geometries: Columns, vline
- Aesthetics: x = difference in percentage, y = expense category, fill = positive/negative

```
expense_breakdown = d3_instLevel %>%
  # only include nwc teams
  filter(institution_name %in% nwc | institution_name == "Willamette University") %>%
  mutate(
    hc_salaries = # calculating the cumulative head coach salaries paid
      coalesce(HDCOACH_SALARY_MEN * NUM_HDCOACH_MEN, 0) + #Making 0 if NA
      coalesce(HDCOACH_SALARY_WOMEN * NUM_HDCOACH_WOMEN, 0) +
      coalesce(HDCOACH_SALARY_COED * NUM_HDCOACH_COED, 0),

    ac_salaries = # calculating the cumulative ac salaries paid
      coalesce(ASCOACH_SALARY_MEN * NUM_ASCOACH_MEN, 0) +
      coalesce(ASCOACH_SALARY_WOMEN * NUM_ASCOACH_WOMEN, 0) +
      coalesce(ASCOACH_SALARY_COED * NUM_ASCOACH_COED, 0),

    recruiting = # calculating the cumulative amount spent on recruiting
      coalesce(RECRUITEXP_MEN, 0) +
      coalesce(RECRUITEXP_WOMEN, 0) +
      coalesce(RECRUITEXP_COED, 0),

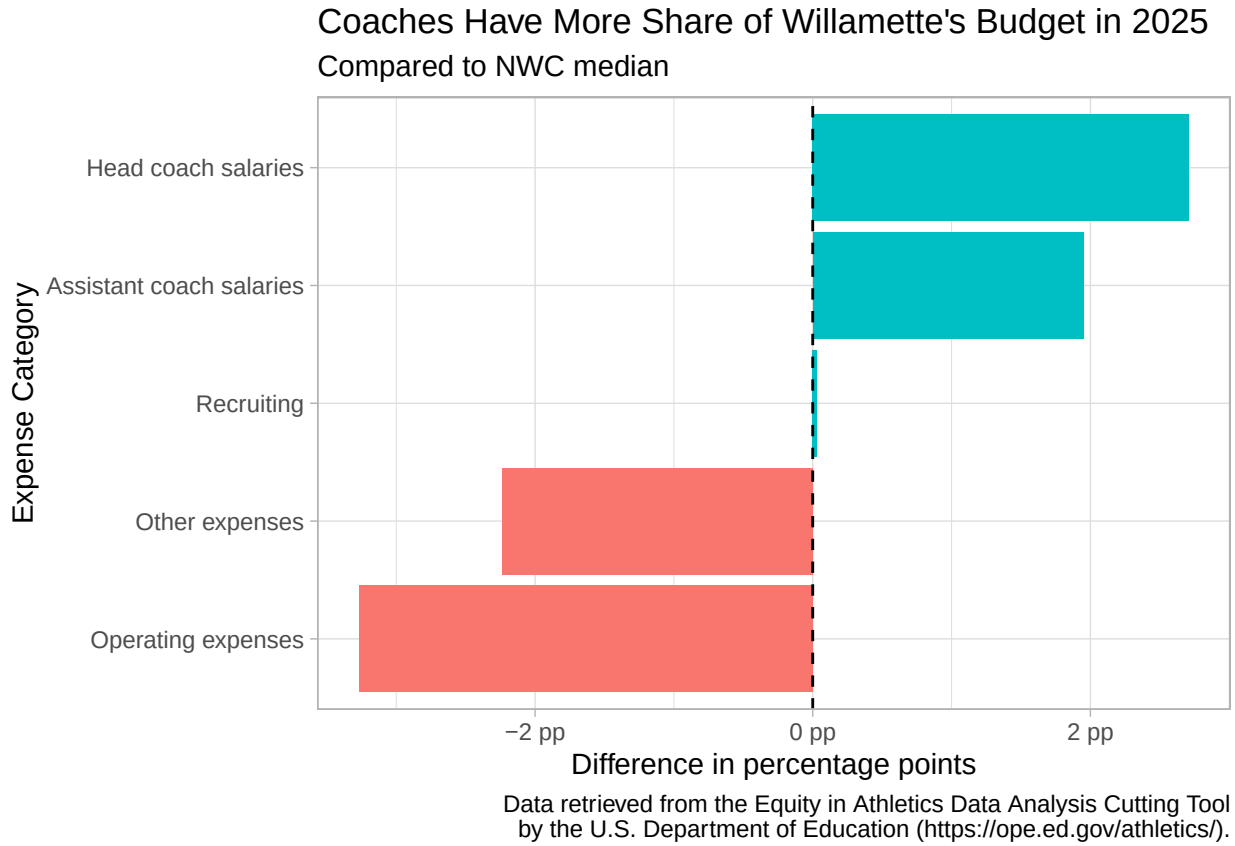
    operating = IL_TOTAL_OPEXP_INCLCOED, # total operating expenses
```

```
    other = GRND_TOTAL_EXPENSE - operating - recruiting - hc_salaries - ac_salaries
  )


selected_year = 2025 #User chooses
title = paste0("Coaches Have More Share of Willamette's Budget in ",
               selected_year) # title that changes with selection

expense_breakdown %>%
  filter(year == selected_year) %>%
  mutate( # calculating percentages
    operating = operating / GRND_TOTAL_EXPENSE * 100,
    recruiting = recruiting / GRND_TOTAL_EXPENSE * 100,
    hc_salaries = hc_salaries / GRND_TOTAL_EXPENSE * 100,
    ac_salaries = ac_salaries / GRND_TOTAL_EXPENSE * 100,
    other = other / GRND_TOTAL_EXPENSE * 100
  ) %>%
  select(
    institution_name,
    operating,
    recruiting,
    hc_salaries,
    ac_salaries,
    other
  ) %>%
  pivot_longer( # making into category/percetage cols
    cols = -institution_name,
    names_to = "category",
    values_to = "percentage"
  ) %>%
  group_by(category) %>%
  mutate( # making the median of each category for all nwc
    nwc_median = median(
      percentage[institution_name %in% nwc],
      na.rm = TRUE
    )
  ) %>%
  ungroup() %>%
  filter(institution_name == "Willamette University") %>% # focus on willamette
  mutate( # make difference category for cols on plot
    difference = percentage - nwc_median,
    category = recode( # rename for plot
      category,
      operating = "Operating expenses",
      recruiting = "Recruiting",
      hc_salaries = "Head coach salaries",
      ac_salaries = "Assistant coach salaries",
      other = "Other expenses"
    )
  ) %>%
  ggplot(aes(x = difference, y = reorder(category, difference),
             fill = difference > 0)) +
  geom_col() +
```

```
  geom_vline(xintercept = 0, linetype = "dashed") + # nwc median line
  scale_x_continuous( # add percentage point labels to x
    labels = \(x) paste0(x, " pp")
  ) +
  theme_light() +
  theme(
    legend.position = "none", # no legend needed
    plot.caption.position = "plot" #Moving caption to the right side
  ) +
  labs(
    title = title,
    subtitle = "Compared to NWC median",
    x = "Difference in percentage points",
    y = "Expense Category",
    caption = "Data retrieved from the Equity in Athletics Data Analysis Cutting Tool\nby the U.S. Depa
  )
```



Analysis: While Willamette coaches need to be more invested in like our earlier graph states, the gap between expenses on athletes needs to be closed. As a result, more immediate change can be forecasted by increasing the percentage of our budget we spend on recruiting, operating expenses, and other expenses.

```
expense_breakdown %>%
  filter(year == 2025) %>%
  select(institution_name, operating, recruiting, hc_salaries, ac_salaries, other) %>%
  pivot_longer(
```

```
    cols = c(operating, recruiting, hc_salaries, ac_salaries, other),
    names_to = "category",
    values_to = "expense"
  ) %>%
  ggplot(aes(
    x = institution_name,
    y = expense,
    fill = category
  )) +
  geom_col() +
  labs(
    x = NULL,
    y = "Grand Total Expenses ($)",
    fill = "Expense Category"
  ) +
  theme_light() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )


expense_breakdown %>%
  filter(year == 2025) %>%
  select(
    institution_name,
    operating, recruiting,
    hc_salaries, ac_salaries, other
  ) %>%
  pivot_longer(
    cols = c(operating, recruiting, hc_salaries, ac_salaries, other),
    names_to = "category",
    values_to = "expense"
  ) %>%
  ggplot(aes(
    x = reorder(institution_name, expense),
    y = expense,
    fill = institution_name == "Willamette University"
  )) +
  geom_col() +
  facet_wrap(~ category, scales = "free_y") +
  coord_flip() +
  scale_fill_manual(
    values = c("TRUE" = "darkred", "FALSE" = "gray70"),
    guide = "none"
  ) +
  labs(
    x = NULL,
    y = "Expenses ($)",
    title = "Athletic Expense Breakdown by School, 2025"
  ) +
  theme_light()
```

```r
expense_breakdown %>%
  filter(year == 2025) %>%
  mutate(
    operating_pct = operating / GRND_TOTAL_EXPENSE * 100,
    recruiting_pct = recruiting / GRND_TOTAL_EXPENSE * 100,
    hc_salaries_pct = hc_salaries / GRND_TOTAL_EXPENSE * 100,
    ac_salaries_pct = ac_salaries / GRND_TOTAL_EXPENSE * 100,
    other_pct = other / GRND_TOTAL_EXPENSE * 100
  ) %>%
  select(
    institution_name,
    operating_pct, recruiting_pct,
    hc_salaries_pct, ac_salaries_pct, other_pct
  ) %>%
  pivot_longer(
    cols = ends_with("_pct"),
    names_to = "category",
    values_to = "percentage"
  ) %>%
  ggplot(aes(
    x = institution_name,
    y = percentage,
    fill = category
  )) +
  geom_col() +
  labs(
    x = NULL,
    y = "Percentage of Grand Total Expenses",
    fill = "Expense Category",
    title = "Athletic Expense Composition by School",
    subtitle = "2025"
  ) +
  scale_y_continuous(
    labels = \(x) paste0(x, "%"),
    limits = c(0, 100)
  ) +
  theme_light() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )


expense_breakdown %>%
  filter(year == 2025) %>%
  select(
    institution_name,
    operating, recruiting,
    hc_salaries, ac_salaries, other
  ) %>%
  pivot_longer(
    cols = -institution_name,
    names_to = "category",
    values_to = "percentage"
  ) %>%
```

```
  mutate(
    category = recode(
      category,
      operating = "Op Exp",
      recruiting = "Rec",
      hc_salaries = "HC Sal",
      ac_salaries = "AC Sal",
      other = "Other"
    )
  ) %>%
  group_by(institution_name) %>%
  mutate(
    category = reorder(category, percentage)
  ) %>%
  ungroup() %>%
  ggplot(aes(x = percentage, y = category)) +
  geom_col(fill = "gray65") +
  facet_wrap(~ institution_name, scales = "free_y") +
  labs(
    x = "Percent of Grand Total Expenses",
    y = NULL,
    title = "Athletic Expense Composition by School",
    subtitle = "2025"
  ) +
  theme_light()

expense_breakdown %>%
  mutate(
    operating_pct = operating / GRND_TOTAL_EXPENSE * 100
  ) %>%
  group_by(year) %>%
  mutate(
    nwc_median = median(operating_pct[institution_name %in% nwc], na.rm = TRUE)
  ) %>%
  ungroup() %>%
  filter(institution_name == "Willamette University") %>%
  ggplot(aes(x = year)) +
  geom_line(aes(y = operating_pct, color = "Willamette University"), linewidth = 1) +
  geom_line(aes(y = nwc_median, color = "Other NWC Schools"), linewidth = 1, linetype = "dashed") +
  labs(
    title = "Operating Expenses as a Share of Athletics Spending",
    x = "Year",
    y = "Operating expenses (% of total)",
    color = "Institution"
  ) +
  scale_color_manual(
    values = c(
      "Other NWC Schools" = "grey60",
      "Willamette University" = "darkred"
    )
  ) +
  scale_y_continuous(
    labels = \(x) paste0(x, "%")
  ) +
```

```
scale_x_continuous(
  breaks = seq(2000, 2025, by = 1)
) +
theme_light()
```